

INVITED TALK / ABSTRACT

Towards Yau's Conjecture: a new estimate for the first eigenvalue of the laplacian of minimal hypersurfaces

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Abstract

The first goal of this talk is to give an overview of the state of the art of the Yau's conjecture, which states that the first eigenvalue of the laplacian of a closed embedded minimal hypersurface in the unit sphere S^{n+1} must be equal to n . Until now this is only known for special cases of minimal hypersurfaces, so we also present a new lower bound of the first eigenvalue that improves previous results and can be extended to more general ambients. If time permits, we will also comment about the case of isoparametric hypersurfaces and the case when the ambient is a submersion of the sphere. This is a joint work with Calos Tapia and Detang Zhou.

Keywords: Yau's conjecture; first eigenvalue; Laplacian; minimal hypersurfaces; unit sphere; isoparametric hypersurfaces; submersion of the sphere